



RTL Design Sherpa

CDC Counter Display Nexys A7 Project Guide 0.90

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1 Document Information

This is the operator / developer guide for the Nexys A7 **CDC Counter Display** project (projects/fpga-systems/NexysA7/cdc_counter_display). It explains what the project is, how it works, how to build / simulate / program / run it, and how the UART characterization harness is configured. It is a practical guide, not a formal architecture specification.

The project ships in two phases that coexist in the same tree:

- **Phase 1** — a standalone, button-driven CDC counter (cdc_counter_display_top).
 - **Phase 2** — a UART-controlled, four-counter CDC demonstrator (cdc_demo_top) with a host program that runs byte-for-byte identically on the FPGA and in a cocotb simulation. This guide focuses on Phase 2.
-

1.1 References

Document references

Source	Title
RTL Design Sherpa	projects/fpga-systems/NexysA7/cdc_counter_display/README.md
RTL Design Sherpa	docs/HARNESS.md (CSR map + wiring)
RTL Design Sherpa	docs/RUNBOOK.md (operator procedures)
RTL Design Sherpa	bin/TBClasses/harness/ (shared UART-char collateral)
Digilent	Nexys A7 Reference Manual
ARM	AMBA AXI and APB Protocol Specifications
.	.

1.2 Conventions

- **Registers are accessed by name**, never by hardcoded offset (see Chapter 4).
- **Clock:** sys_clk = 100 MHz on-board oscillator. Per-counter source clocks (ctr_clk[i]) are asynchronous to sys_clk and to each other.

- **Reset:** CPU_RESETN (board button BTNR, active-low) or CTRL.soft_reset.
 - **Register/signal prefixes:** r_ registered, w_ combinational.
-

1.3 Revision History

Revision history

Version	Date	Notes
0.90	2026-07-18	Initial project guide

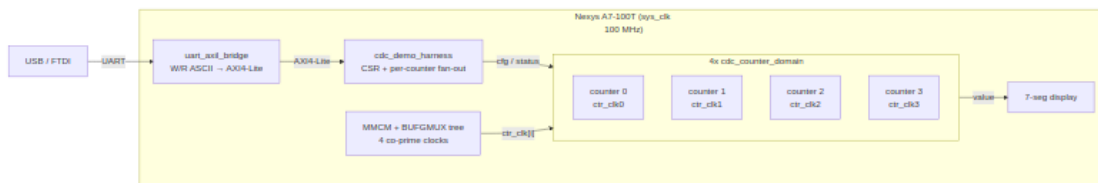
2 Overview

2.1 What it is

CDC Counter Display is an educational Nexys A7 project that demonstrates **clock domain crossing (CDC)** — both done correctly and broken on purpose — on real silicon. Four independent counters each run in their own asynchronous clock domain; a host program over UART configures them, reads their values back across the CDC boundary, and can deliberately select an unsafe crossing to *watch CDC fail* on the 7-segment display.

Phase 2 (`cdc_demo_top`) is wired on the standard Nexys A7 characterization **spine**: a single USB-UART link feeds a `uart_axil_bridge` that converts an ASCII W/R command protocol into AXI4-Lite, which drives the `cdc_demo_harness` CSR block. The harness fans configuration out to four `cdc_counter_domain` instances and collects their CDC'd status back.

2.1.1 Figure 1.1: CDC Demo Harness Spine



CDC demo harness spine

Source: [01_harness_spine.mmd](#)

2.2 What it demonstrates

- **Safe multi-bit CDC.** With a Gray-coded or FIFO-based crossing, the host reads coherent counter values at any source-clock speed.
- **Broken CDC, visibly.** Put a counter in NO-CDC mode with auto-increment, then sweep its source clock from slow to fast: the display stays clean at low speeds and visibly scrambles at high speeds (multi-bit bus skew), while the other counters — in a safe mode — stay clean the whole time.
- **Five selectable CDC strategies** per counter: NO-CDC, stretch, sync-FIFO, two-phase handshake, four-phase handshake.
- **Sim / silicon equivalence.** The same host program drives the FPGA and a cocotb simulation over the identical UART byte stream (Chapter 6).

2.3 What you need

- Nexys A7-100T board (XC7A100T-1CSG324C) + USB cable.
- Xilinx Vivado (for make bitstream / make program).
- A Python environment sourced from env_python (for the host CLI and sim).

2.4 Where things live

Project file layout

Path	Contents
build-demo/rtl/ cdc_demo_top.sv	Board top: clocking tree, buttons, harness, display
build-demo/rtl/ cdc_demo_harness.sv	AXI4-Lite CSR + per-counter fan-out
build-demo/rtl/ cdc_counter_domain.sv	One counter + its CDC paths (5 modes)
build-demo/rtl/ cdc_demo_csr.rdl	Register descriptor (generates the by-name regmap)
build-demo/host/	cdc_demo.py driver, cdc_programs.py, host_cdc_demo.py CLI
build-demo/dv/	UART-equivalence cocotb sim (tb, tests, filelist, regmap)
Makefile	Build / sim / program / host targets

3 How It Works

3.1 Top-level structure (cdc_demo_top)

The board top instantiates, from UART inward:

1. **uart_axil_bridge** — decodes the ASCII W <addr> <data> / R <addr> protocol into single-beat AXI4-Lite. CLKS_PER_BIT is derived from $\text{SYS_CLK_HZ} / 115200$.
2. **cdc_demo_harness** — the AXI4-Lite CSR slave (Chapter 5). Stores per-counter config, exposes CDC'd status, and generates single-cycle pulses (CFG_LOAD, HOST_PRESS).
3. **Clocking tree** — an MMCME2_BASE produces four co-prime divided clocks; per counter a BUFMUX_CTRL tree glitchlessly selects one of five sources.
4. **Four cdc_counter_domain instances** — the counters and their CDC paths.
5. **Display** — DISP_SELECT picks which counter's value drives the 7-seg.

3.2 Clocking and the four source clocks

The MMCM synthesizes four **mutually asynchronous** clocks from the 100 MHz reference using pairwise co-prime divisors, plus a per-counter sys_clk-derived divided clock:

Per-counter clock sources

clock_select	Source	Frequency
0	MMCM CLKOUT0 (÷11)	72.7 MHz
1	MMCM CLKOUT1 (÷29)	27.6 MHz
2	MMCM CLKOUT2 (÷67)	11.9 MHz
3	MMCM CLKOUT3 (÷128)	6.25 MHz
4	clock_divider (uses div_pickoff)	$\text{sys_clk} / 2^{(\text{pickoff}+1)}$

The co-prime divisors {11, 29, 67, 128} mean no two outputs share an edge alignment for a very long time — for CDC purposes, truly asynchronous. The divided-clock branch (select 4) keeps

a slow, visibly-counting rate available for the “watch it count” demo; `div_pickoff = 23` gives roughly 6 Hz.

3.3 One counter domain (`cdc_counter_domain`)

Each counter lives entirely in its `ctr_clk[i]` domain and crosses two ways:

- **sys_clk → ctr_clk (config in):** INIT / INCREMENT as quasi-static 2-FF synchronized buses; CFG_LOAD / HOST_PRESS as single-shot pulses through a `sync_pulse` module.
- **ctr_clk → sys_clk (status out):** VALUE, PRESS_COUNT, CTR_CLK_TICKS crossed back for readback. **The VALUE path is where CDC_MODE selects the crossing strategy** — this is the heart of the demo.

On each `ctr_clk` edge the counter loads INIT on a CFG_LOAD pulse, otherwise advances by INCREMENT when a press event occurs. A press event is a debounced button, a HOST_PRESS pulse, or — when `AUTO_INC = 1` — every `ctr_clk` edge.

3.4 The five CDC modes (CDC_MODE)

The five CDC modes for the VALUE readback path

Mode	Name	Primitive	Safe?
0	NO-CDC	raw flop per bit	No (multi-bit skew at fast clocks)
1	STRETCH	<code>cdc_open_loop</code> (pulse stretch + sync)	Up to a tuned cliff (~25 MHz)
2	SYNC-FIFO	<code>fifo_async</code> (Gray pointers)	Yes
3	TWO-PHASE	<code>cdc_2_phase_handshake</code>	Yes
4	FOUR-PHASE	<code>cdc_4_phase_handshake</code>	Yes

Mode 0 (NO-CDC) samples the multi-bit value with plain flops — at a fast source clock the bits arrive skewed and the host reads transient garbage. That is the intended failure. In Verilator (no metastability model) mode 0 reads the true value deterministically, which is why the simulation asserts on mode 0 for exact values (Chapter 6) but the *board* shows scramble at speed.

3.5 Display

DISP_SELECT chooses which counter's 16-bit VALUE drives the 7-segment digits; the upper digits show the selected counter's CDC_MODE and clock-select so the operator can read the current configuration off the board.

4 Test Methodology

This chapter describes what is exercised on the FPGA block (Figure 2.1) and how the results are judged. CDC Counter Display is a **demonstration**, so the “measurement” is a coherence test of the clock-domain-crossing readback rather than a throughput sweep.

4.1 What is under test

The four `cdc_counter_domain` value-readback paths, each selectable among five CDC strategies (CDC_MODE 0–4) and driven by an asynchronous source clock whose rate the host can sweep. The question under test: **does the multi-bit counter value cross into `sys_clk` coherently?**

4.2 What is measured, and how

What is measured

Quantity	How it is obtained
Value coherence	Read VALUE by name N times; compute min / max / number of unique samples
Press determinism	After count HOST_PRESS pulses, check VALUE == INIT + count*INCREMENT and PRESS_COUNT delta
Reload	CFG_LOAD reloads VALUE to INIT without disturbing PRESS_COUNT
Mode selection	Each CDC_MODE code round-trips through the CSR and selects a distinct path

A run configures a counter (mode, init, increment, clock), injects stimulus (HOST_PRESS or AUTO_INC), then samples VALUE/PRESS_COUNT over the by-name bridge. In the “watch-fail” workload the counter is put in NO-CDC with AUTO_INC = 1 and its source clock is swept slow→fast; the **spread and unique count of the VALUE samples** is the metric — a few unique values means clean crossing, a wide 0x00–0xFF spread means multi-bit skew.

4.3 Workloads

Demonstration workloads

Workload	Purpose
press	Deterministic increment check (safe CDC mode)
cfg_load	Reload semantics
cdc_mode	All five mode codes round-trip

Workload	Purpose
watch_fail	Sweep an unsafe crossing slow→fast and watch it break

4.4 The oracle, and what sim can/can't show

The oracle is the arithmetic expectation ($\text{INIT} + \text{presses} * \text{INCREMENT}$) for the safe modes. In Verilator the NO-CDC path reads the *true* value (no metastability model), so the sim asserts exact values even in mode 0 — the analog “garbage” is a silicon-only effect. That asymmetry is the point of the demo, not a defect (Chapter 6, Sim Equivalence).

4.5 Measurement pitfalls

- NO-CDC garbage at speed is **expected** — use a safe CDC_MODE (2/3/4) before asserting exact values.
- PRESS_COUNT always uses Gray-coded CDC and stays coherent regardless of mode; prefer it when you need a trustworthy count.

5 Build and Run

All commands are run from `projects/fpga-systems/NexysA7/cdc_counter_display/` after sourcing the Python environment:

```
cd /path/to/RTLDesignSherpa
source env_python           # sets SIM=verilator, PATH, PYTHONPATH,
                             REPO_ROOT
cd projects/fpga-systems/NexysA7/cdc_counter_display
```

5.1 The workflow at a glance

```
make regmap -> make consistency -> make sim          (prove it in
simulation)
                                     |
                                     v
                                make bitstream -> make program -> make host-cdc_demo
(on the board)
```

Targets go to `build-demo` by default; add `BUILD=phase1` for the button-only build. All flow logic lives in the global `make/fpga_flow.mk` (`make targets` lists what was discovered).

5.2 Make targets

5.2.1 Simulation

Simulation make targets

Target	What it does
<code>make sim BUILD=phase1</code>	Phase-1 CocoTB sim of <code>cdc_counter_display_top</code> .
<code>make sim</code>	UART-equivalence sim: wraps the real <code>uart_axil_bridge</code> + harness and runs the host programs over a cocotb UART master (<code>build-demo/dv/tests/test_cdc_demo_uart.py</code>). Four tests: <code>smoke</code> , <code>press</code> , <code>cfg_load</code> , <code>cdc_mode</code> .

5.2.2 Register collateral

Register-collateral make targets

Target	What it does
<code>make regmap</code>	Regenerate <code>build-demo/dv/tbclasses/cdc_demo_csr_regmap.py</code> from <code>build-demo/rtl/cdc_demo_csr.rdl</code> . Run after editing the RDL — never hand-edit the

Target	What it does
<code>make consistency</code>	regmap. Guard test: the generated regmap must match the hand-written <code>cdc_demo_harness.sv</code> (offsets + per-counter block).

5.2.3 Bitstream

Bitstream make targets

Target	What it does
<code>make bitstream</code>	Build the demo bitstream <code>build-demo/fpga/bitstream/cdc_demo.bit</code> with Vivado (~5–10 min). <code>make project</code> creates the Vivado project only; <code>make synth</code> stops after synthesis.
<code>make program</code>	Flash the board (shared board layer, registry-pinned JTAG serial). <code>make ports / make board-info</code> inspect the board.
<code>make lint</code>	Verilator lint of the build's filelist closure (Xilinx clocking primitives are stubbed — see Chapter 7).
<code>make utilization / make timing</code>	Print the latest report summaries.
<code>make keep</code>	Promote the current bitstream + reports out of the clean blast radius.

5.3 The host CLI (`build-demo/host/host_cdc_demo.py`)

The CLI builds a `CdcDemoDriver`, resolves the serial port through the shared board layer (narrows to this board's ports by USB serial, then probes for the CDC1 build ID unless `--port` is given), and dispatches to the authored-once programs in `cdc_programs.py`. Run it as `make host-cdc_demo ARGS=...` or directly:

```
# Verify the link and dump all four counters' defaults
make host-cdc_demo ARGS=smoke
```

```
# Inject 1000 host presses to counter 2; checks VALUE = INIT + 1000*INCREMENT
make host-cdc_demo ARGS="press --counter 2 --count 1000"
```

```
# CFG_LOAD reload check / CDC_MODE round-trip
```

```

make host-cdc_demo ARGS="cfg-load --counter 1"
make host-cdc_demo ARGS="cdc-mode --counter 0"

# Real-time monitor of all four counters
make host-cdc_demo ARGS=monitor

# The headline demo: NO-CDC + auto-increment, sweep the clock slow ->
fast
make host-cdc_demo ARGS="watch-fail --counter 2"

# Force a specific port instead of autodetect (direct invocation)
python3 build-demo/host/host_cdc_demo.py --port /dev/ttyUSB1 smoke

```

Each subcommand prints a human-readable result and returns a non-zero exit code on failure, so the CLI doubles as a board bring-up smoke test.

5.3.1 The “watch it fail” procedure

1. Program the board (make program).
2. `make host-cdc_demo ARGS="watch-fail --counter 2"` sets counter 2 to NO-CDC + auto-increment, sets DISP_SELECT to 2, and sweeps div_pickoff from slow to fast, sampling VALUE at each step.
3. Read the on-board 7-seg: clean counting at slow pickoffs, visible scramble at fast pickoffs. Compare against a counter left in a safe mode — it stays clean.

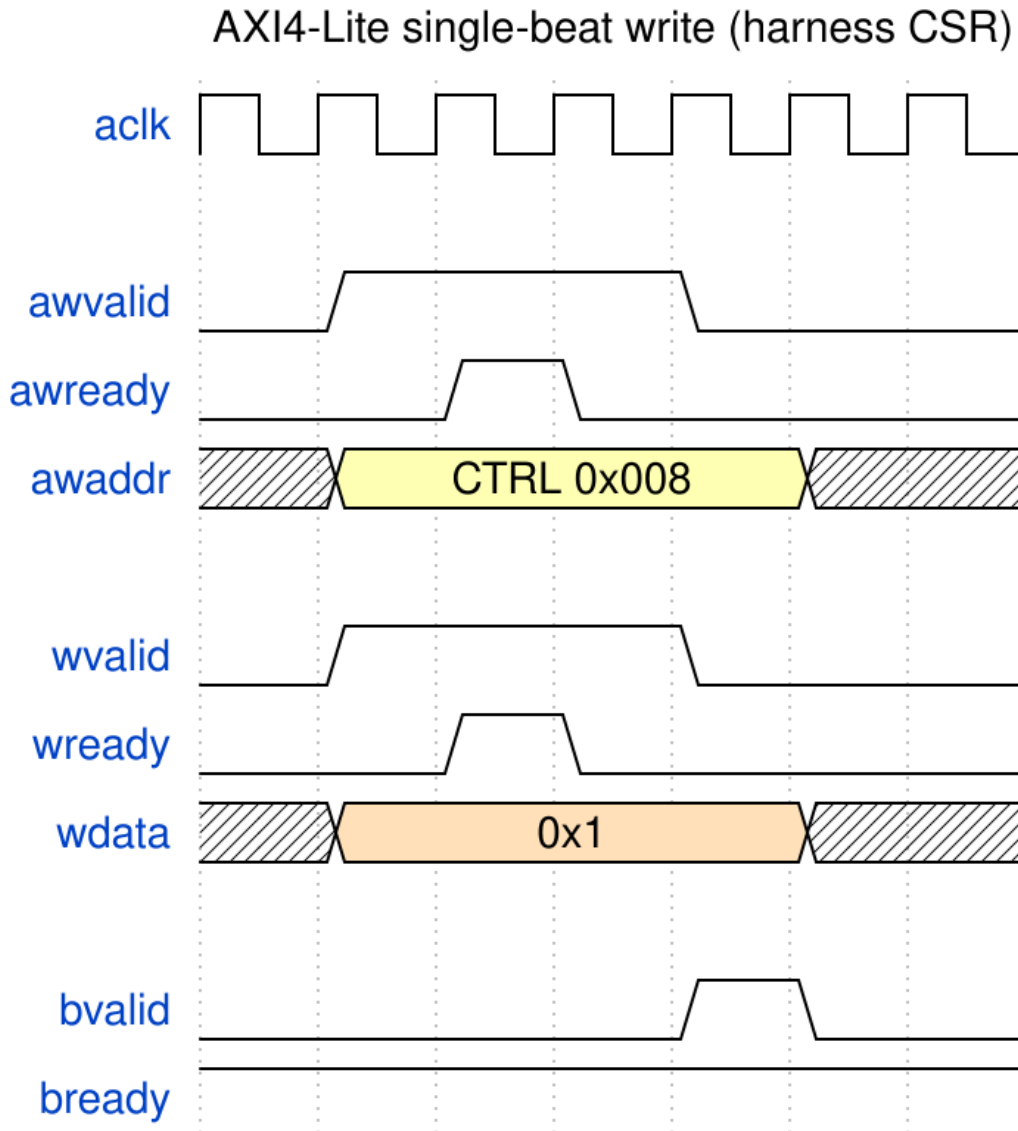
6 Harness CSR Configuration

The `cdc_demo_harness` is a single flat AXI4-Lite slave. The host reaches it through the UART bridge at **harness base 0x0**. Every register is 32 bits, word-addressable.

Registers are accessed by name, never by offset. The layout is authoritative in `rtl/cdc_demo_harness.sv`, described in `rtl/cdc_demo_csr.rdl`, and compiled to `dv/tbclasses/cdc_demo_csr_regmap.py` (PeakRDL). The driver, the host programs, and the sim all resolve names through that regmap; the consistency test guards the three against drift.

Every named write becomes a single AXI4-Lite beat that the UART bridge drives into the harness (a named read is the mirror AR/R exchange):

6.0.0.1 Waveform 5.1: AXI4-Lite Single-Beat Write



AXI4-Lite single-beat write

Source: [01_axil_write.json](#)

6.1 Global registers

Global CSR registers — register / field / offset / bit slice

Register	Offset	Field	Bits	Access	Description
BUILD_ID	0x000	value	[31:0]	R	ASCII “CDC1”

Register	Offset	Field	Bits	Access	Description
					(0x43434331) — read first to confirm the bitstream
STATUS	0x004	alive0–alive3	[3:0]	R	Counter <i>i</i> toggled within the last ~1 s window
STATUS	0x004	uart_rx	[4]	R	UART RX activity
STATUS	0x004	uart_tx	[5]	R	UART TX activity
STATUS	0x004	any_written	[6]	R	Any CSR written since reset
STATUS	0x004	reset_ok	[31]	R	Reset deasserted / harness alive
CTRL	0x008	soft_reset	[0]	RW	Pulse: full sys_clk reset
CTRL	0x008	freeze	[1]	RW	Freeze all counters
CTRL	0x008	ignore_btn	[2]	RW	Ignore physical buttons (host-press only)
DISP_SELECT	0x00C	sel	[1:0]	RW	Which counter drives the 7-seg
SCRATCH	0x010	value	[31:0]	RW	Link-sanity scratch (write, read back)

6.2 Per-counter block

Counter *i* (0–3) occupies $0x040 + i*0x40$. The regmap keys are COUNTER{*i*}_<REG> (e.g. COUNTER2_DIVISOR).

*Per-counter CSR block — counter *i* absolute offset = $0x40 + i*0x40 + \text{column}$*

Register	+Offset	Field	Bits	Access	Description
DIVISOR	+0x00	clock_select	[2:0]	RW	Clock source (0–3 MMCM, 4 =

Register	+Offset	Field	Bits	Access	Description
DIVISOR	+0x00	div_pic koff	[12:8]	RW	divided) Divided-clock pickoff (when clock_select=4)
INIT	+0x04	value	[15:0]	RW	Value loaded by CFG_LOAD
INCREMENT	+0x08	value	[15:0]	RW	Advance amount per press
CFG_LOAD	+0x0C	strobe	[0]	W	Pulse: load INIT into the counter
HOST_PRESS	+0x10	strobe	[0]	W	Pulse: inject one virtual press
VALUE	+0x14	value	[15:0]	R	Current value, CDC'd per CDC_MODE
PRESS_COUNT	+0x18	value	[15:0]	R	Debounced press count (always Gray-CDC'd)
CTR_CLK_TIC KS	+0x1C	value	[31:0]	R	Free-running ctr_clk tick count
CDC_MODE	+0x20	mode	[2:0]	RW	CDC strategy for VALUE (0–4; see Chapter 2)
AUTO_INC	+0x24	en	[0]	RW	1 = advance every ctr_clk edge

6.3 By-name access

The driver (`host/cdc_demo.py`) wraps `UartRegisterMap`. Whole-register and individual-field access both work:

```

from cdc_demo import CdcDemoDriver
d = CdcDemoDriver(port="/dev/ttyUSB1")           # or bridge=... for
sim                                              # -> 0x43434331

d.build_id()
```

```

d.scratch(0xDEADBEEF)           # write + read back
d.disp_select(2)                 # DISP_SELECT.sel = 2

c = d.counter(2)
c.set_cdc_mode(2)                #
COUNTER2_CDC_MODE.mode = 2
c.set_div_pickoff(23)            # DIVISOR.
{clock_select=4, div_pickoff=23}
c.set_init(0x10); c.set_increment(3); c.load() # load INIT
c.press()                        # HOST_PRESS
c.value(), c.press_count()       # CDC'd status
readback

```

Packed fields are spliced by name with a read-modify-write, so setting `div_pickoff` never disturbs `clock_select`. Write-only strobes (`CFG_LOAD`, `HOST_PRESS`) are writable by name but read back as 0 by design.

6.4 Regenerating the regmap

`rtl/cdc_demo_harness.sv` is hand-written; the RDL is a descriptor of it. After editing either the SV or the RDL:

```

make regmap      # regenerate dv/tbclasses/cdc_demo_csr_regmap.py
from the RDL
make consistency # assert regmap == SV (globals + reconstructed per-
counter)

```

Never hand-edit the generated regmap — it is regenerated from the RDL.

7 Simulation / Silicon Equivalence

This project follows the shared Nexys A7 UART-characterization methodology (bin/TBClasses/harness/): **one authored-once host program runs byte-for-byte identically on the FPGA and in a cocotb simulation.** The equivalence boundary is the ASCII W/R byte stream on the UART wire — if the same program emits the same bytes, the same `uart_axil_bridge` RTL decodes them the same way on silicon and in sim.

7.1 The layered stack (only the bottom layer differs)

Programs	<code>cdc_programs.py</code>	authored once, unchanged
FPGA/sim		
Driver	<code>CdcDemoDriver + UartRegisterMap</code>	by-name registers
Protocol	<code>UARTAxiBridge (W/R ASCII)</code>	same code both sides
--	byte stream	

Channel	<code>SerialChannel (pyserial)</code>	<code>CocotbUartChannel (cocotb)</code>
Wire	<code>FTDI/USB UART</code>	<code>UARTMaster/Monitor on DUT pins</code>

The bridge is **injectable**: `CdcDemoDriver(port=...)` opens a real serial port; `CdcDemoDriver(bridge=UARTAxiBridge(channel=cocotb_channel))` drives the sim. Nothing above the wire forks.

7.2 How the sim runs the same program

`dv/tb/cdc_demo_uart_tb_top.sv` wraps the **real** `uart_axil_bridge + cdc_demo_harness + four cdc_counter_domain` instances. `dv/tests/test_cdc_demo_uart.py`:

1. Starts `acclk` and reset, and drives the four `ctr_clk[i]` at co-prime periods.
2. Builds a `CocotbUartChannel` via `make_uart_channel(dut, dut.acclk, CLKS_PER_BIT)`.
3. Constructs `CdcDemoDriver(bridge=UARTAxiBridge(channel=chan))`.
4. Runs the unmodified `cdc_programs.*` functions inside `cocotb.external(...)`.

The synchronous host program runs in a worker thread; the channel's read/write are `cocotb`. function wrappers that drive the UART master and advance simulation time. The four `cocotb` tests (`smoke`, `press`, `cfg_load`, `cdc_mode`) assert the same results the CLI checks on the board.

7.3 What the sim can and cannot reproduce

- **CAN:** everything digital — the bridge RTL, CSR decode, per-counter CDC datapaths, register logic. The sim proves the byte protocol and the harness contract before you ever touch the board.

- **CANNOT:** the analog clock tree. `cdc_demo_top`'s `MMCME2_BASE` / `BUFGMUX_CTRL` / `IBUF` / `BUFG` do not simulate in Verilator, so the `tb_top` drives `ctr_clk[i]` behaviorally instead. Consequently the NO-CDC “garbage read” is a silicon-only effect — in Verilator (no metastability) mode 0 reads the true value, so the sim asserts exact values in NO-CDC while the board shows scramble at speed.

7.4 Gotchas worth keeping

- **Lower `CLKS_PER_BIT` in sim** (16, in both the `tb_top` param and the BFM) — bit-timing only, the byte stream is unchanged, so equivalence holds.
- **Wrap the full harness, not the inner block** — the real `uart_axil_bridge` + `cdc_demo_harness` must be in the DUT, or the two flows are only “vaguely similar,” not equivalent.
- **Use `cocotb.function`, not a free-running pump** — a pump coroutine plus thread queues stalls the scheduler; `cocotb.function` is what steps the sim from worker-thread calls.

8 Troubleshooting

8.1 The host CLI cannot find the board

The USB-UART re-enumerates across reboots and replugs, so the port number drifts. The CLI defaults to `--port auto`, which resolves the port through the shared board layer: it narrows to this board's ports by USB serial (the board registry in `projects/fpga-systems/bin/boards/` knows the FTDI serial) and keeps the one that answers `BUILD_ID == 0x43434331` ("CDC1"). If autodetect fails:

- Confirm the board is powered and programmed with the **demo** bitstream (`make program`), not `build-phase1`.
- Run `make ports` to see which `ttyUSB` this board is on right now, and `make board-info` for the registry's serial. A swapped board unit shows up as "no UART ports found" quoting a serial that is no longer attached.
- Check the FTDI cable and that no other program holds the port.
- Force a specific device with `--port /dev/ttyUSB1` (still probed, so a stale path fails loudly instead of driving the wrong board).

8.2 Values look scrambled / never settle

Expected in **NO-CDC mode at a fast clock** — that is the demonstration. For coherent reads, put the counter in a safe `CDC_MODE` (2 = `SYNC-FIFO`, or 3/4 for the handshakes) before reading `VALUE`. `PRESS_COUNT` always uses Gray-coded CDC and is coherent regardless of mode.

8.3 `make lint` — Xilinx primitive errors

`cdc_demo_top` instantiates `MMCME2_BASE` / `BUFGMUX_CTRL` / `IBUF` / `BUFG`, which Verilator cannot find. The shared stub file (`projects/components/utility-ip/misc/rtl/verilator_xilinx_stubs.sv`, pulled in through its filelist by `build-demo/rtl/filelists/cdc_demo_top.f`) provides ``ifdef VERILATOR`-guarded pass-through stubs (Vivado uses the real unisims at synthesis; the `create_project tcl` drops the stub file from the project). If lint fails with "Cannot find module" for one of these, confirm the stub filelist is still included in the build's `.f` closure.

8.4 `make sim BUILD=phase1` fails to compile

Phase 1 uses `clock_divider.sv`, which relies on the ``ALWAYS_FF_RST` macro from `rtl/amba/includes/reset_defs.svh`. The build's filelist (`build-phase1/rtl/filelists/cdc_counter_display_top.f`) carries the `+incdir` and the `reset-defs` sub-filelist; the test resolves its sources from that same closure. If you see "Define or directive not defined: `ALWAYS_FF_RST`", that include entry is missing from the filelist.

8.5 make consistency fails after editing registers

The generated regmap drifted from the hand-written SV. Regenerate and re-check:

```
make regmap  
make consistency
```

If it still fails, the SV header table or the CTR_OFF_* localparams in `cdc_demo_harness.sv` disagree with `build-demo/rtl/cdc_demo_csr.rdl` — reconcile the three.

8.6 Sim runs but nothing advances

If a UART-equivalence test hangs, the transport is almost certainly using a pump instead of `cocotb.function`, or `CLKS_PER_BIT` was left at the silicon value (868), making each command hundreds of thousands of clocks. Both are covered in Chapter 6.